

Student Academic Management System

AI-Powered NCERT RAG Academic Platform

Detailed Technical & Functional Documentation

1. Introduction

The Student Academic Management System is an AI-powered educational platform designed to modernize academic learning and school management using:

- RAG (Retrieval-Augmented Generation)
- AI-generated quizzes
- AI-generated flashcards
- Gamified learning
- Academic analytics
- Multi-role school management

The platform primarily uses NCERT textbooks and custom educational content uploaded by schools to create intelligent learning experiences for students.

The system transforms static educational materials into:

- Interactive quizzes
 - Flashcards
 - Performance analytics
 - Personalized learning experiences
 - Gamified learning journeys
-

2. Main Objective of the Project

The main goal of the project is to:

- Digitize academic management
- Reduce teacher workload using AI
- Improve student engagement
- Provide personalized learning
- Automatically generate educational content
- Track student progress continuously
- Create a scalable academic ecosystem for schools

3. Problems Solved by the System

Traditional school systems face several issues:

Existing Problem	Solution in This Project
Manual quiz preparation	AI-generated quizzes
Static learning material	Interactive flashcards & quizzes
Low student engagement	Gamified learning system
Difficult progress tracking	Analytics dashboards
Large syllabus management	AI-powered retrieval system
Repetitive teacher work	Automated educational content generation
No personalized learning	Student-wise performance recommendations

4. User Personas

4.1 Super Admin (Platform Provider)

Who Are They?

The organization/service provider managing the entire platform.

Responsibilities

- Manage all schools
- Monitor platform activity
- Manage subscriptions
- View analytics
- Monitor AI usage
- Configure platform settings

Main Goal

Ensure the entire platform works efficiently across multiple schools.

4.2 School Admin

Who Are They?

School-level administrators.

Responsibilities

- Add teachers
- Add students
- Manage classes and subjects
- Assign teachers to classes
- Monitor school performance

Main Goal

Digitally manage the school's academic structure and provide centralized learning resources for students.

School-Level Global Learning System

The platform includes a centralized school-level learning repository.

School admins can publish:

- Global quizzes
- Global flashcards
- Global practice materials

These resources are accessible to all students within the school for self-learning and practice.

Global Learning Features

Global Flashcards

Students can:

- Access school-wide flashcards
 - Practice by subject/chapter
 - Revise concepts independently
 - Bookmark important flashcards
-

Global Quizzes

Students can:

- Attend self-practice quizzes
- Practice unlimited attempts (configurable)
- View instant reports
- Improve weak subjects

Difference Between Global & Personalized Learning

Type	Created By	Accessible To	Purpose
Global Flashcards/ Quizzes	School Admin / School	Entire school students	Self-learning & practice
Personalized Assignments	Teacher	Specific class/ students	Classroom learning & evaluation

4.3 Teacher

Who Are They?

Subject teachers who manage learning activities.

Responsibilities

- Upload study materials
- Generate personalized quizzes
- Generate personalized flashcards
- Assign assessments
- Monitor student progress
- Analyze student performance

Main Goal

Teach efficiently with AI assistance.

4.4 Student

Who Are They?

Learners using the platform.

Responsibilities

- Attend quizzes
- Learn using flashcards
- Complete assignments
- Track performance
- Participate in leaderboards

Main Goal

Learn interactively and improve academic performance.

5. System Modules Overview

The platform consists of multiple interconnected modules.

Module	Purpose
Authentication Module	Secure login & RBAC
School Management Module	Manage schools/classes/users
File Ingestion Module	Upload and process documents
RAG Pipeline	Retrieve educational knowledge
Flashcard Generation Module	AI-generated flashcards
Quiz Generation Module	AI-generated quizzes
Assignment Module	Quiz assignment workflows
Analytics Module	Performance tracking
Gamification Module	XP, badges, streaks
Dashboard Module	User-specific dashboards

6. High-Level System Architecture

```
Frontend (Next.js)
  ↓
Backend API Layer (FastAPI)
  ↓
Business Logic Layer
  ↓
AI Services Layer
  ↓
RAG Pipeline
  ↓
Vector Database (ChromaDB)
  ↓
PostgreSQL Database
  ↓
Cloud/File Storage
```

7. Technology Stack

7.1 Frontend Technologies

Technology	Purpose
Next.js	Frontend framework
React.js	UI development
Tailwind CSS	Styling
ShadCN UI	UI components
Framer Motion	Animations
Recharts	Analytics charts

7.2 Backend Technologies

Technology	Purpose
FastAPI	Backend APIs

Technology	Purpose
Python	AI & backend logic
SQLAlchemy	ORM
JWT	Authentication
Pydantic	Data validation

7.3 Database Technologies

Technology	Purpose
PostgreSQL	Main relational database
ChromaDB	Vector database

7.4 AI Technologies

Technology	Purpose
LangChain	RAG orchestration
LlamaIndex	Document indexing
Qwen/Llama/Mistral	LLMs
BGE Embeddings	Embedding generation

7.5 Cloud & Storage

Technology	Purpose
AWS S3	File storage
Docker	Containerization
Nginx	Reverse proxy
Redis	Caching/session management

8. Authentication & Authorization Workflow

Purpose

Securely authenticate users and provide role-based access.

Login Workflow

```
User enters email/password
↓
Backend validates credentials
↓
JWT token generated
↓
Role assigned
↓
User redirected to dashboard
```

Roles

Role	Access Scope
Super Admin	Entire platform
School Admin	School-specific data
Teacher	Assigned classes/students
Student	Personal academic data

9. School Management Workflow

Workflow

```
School Admin creates:
- Classes
- Sections
```

```
- Subjects
  ↓
Adds teachers
  ↓
Adds students
  ↓
Maps teachers to subjects/classes
  ↓
Maps students to sections
```

10. Academic Content Ingestion Module

Purpose

Convert educational documents into AI-searchable knowledge.

Supported File Types

- PDF
 - DOCX
 - TXT
-

Upload Workflow

```
Teacher uploads document
  ↓
File parser extracts text
  ↓
Text cleaning performed
  ↓
Content chunked
  ↓
Embeddings generated
  ↓
Embeddings stored in vector DB
  ↓
Metadata stored in PostgreSQL
```

11. Document Processing Pipeline

Step 1 — File Upload

The teacher uploads:

- NCERT textbooks
- Notes
- Worksheets
- Custom material

Step 2 — Text Extraction

Libraries used:

File Type	Library
PDF	PyMuPDF/pdfplumber
DOCX	python-docx
TXT	Native parser

Step 3 — Text Cleaning

Processing includes:

- Removing special characters
- Removing unwanted spaces
- Header/footer removal
- OCR cleanup

Step 4 — Chunking

Why Chunking?

Large documents cannot be directly sent to LLMs.

So documents are divided into smaller meaningful chunks.

Recommended Chunk Strategy

Parameter	Value
Chunk Size	300–500 tokens
Overlap	50–100 tokens
Strategy	Chapter/topic-wise

Step 5 — Embedding Generation

Embeddings convert text into vectors.

Example:

```
"Pressure is force per unit area"  
↓  
[0.234, 0.893, 0.112, ...]
```

Step 6 — Vector Database Storage

Stored Data:

- Embedding vectors
 - Original chunks
 - Metadata
-

12. RAG Pipeline Architecture

What is RAG?

RAG stands for Retrieval-Augmented Generation.

It helps AI generate accurate educational content using uploaded textbooks.

RAG Workflow

```
User requests quiz generation
↓
Relevant chunks retrieved from vector DB
↓
Chunks added as context
↓
LLM receives context
↓
LLM generates accurate educational content
```

Why RAG?

Without RAG:

- AI hallucinates
- Generates generic answers

With RAG:

- AI uses actual textbook content
- Generates syllabus-based output
- Improves accuracy

13. Flashcard Generation Workflow

Workflow

```
Teacher selects:
- Subject
- Chapter
- Difficulty
↓
Backend retrieves relevant chunks
↓
LLM generates flashcards
↓
```

Flashcards validated

↓

Stored in database

Flashcard Types

- Concept flashcards
- Definition flashcards
- Formula flashcards
- Revision cards

Flashcard Example

Front	Back
What is pressure?	Force applied per unit area

14. Quiz Generation Workflow

Workflow

Teacher selects:

- Subject
- Chapter
- Question count
- Difficulty

↓

RAG retrieves educational chunks

↓

LLM generates questions

↓

Backend validates structure

↓

Quiz saved to database

Quiz Types

- MCQs
 - True/False
 - Fill in blanks
-

Quiz Validation

Validation checks:

- Correct answer existence
 - Duplicate removal
 - Option validation
 - Formatting consistency
-

15. Quiz Assignment Workflow

Personalized Teacher Assignments

Teachers can create and assign personalized academic content.

These assignments are:

- Class-specific
- Section-specific
- Student-specific

Workflow

```
Teacher creates/generates quiz
↓
Teacher assigns quiz to class
↓
Sets duration and deadline
↓
Students receive notification
↓
Students attend quiz
```

16. Student Quiz Attempt Workflow

Workflow

```
graph TD; A[Student opens assigned quiz] --> B[Quiz timer starts]; B --> C[Student answers questions]; C --> D[Student submits quiz]; D --> E[Auto evaluation performed]; E --> F[Performance report generated];
```

17. Auto Evaluation System

Features

The backend automatically:

- Compares answers
- Calculates score
- Generates accuracy percentage
- Generates reports

Example

Correct Answers	Score
8/10	80%

18. Gamification Module

Purpose

Increase student engagement and consistency.

Components

Feature	Purpose
XP System	Reward learning
Levels	Learning progression
Streaks	Consistency tracking
Badges	Achievement system
Leaderboards	Competitive learning

XP Workflow

```
graph TD; A[Student completes activity] --> B[XP calculated]; B --> C[XP added to profile]; C --> D[Leaderboard updated]; D --> E[Level recalculated];
```

Example XP Rules

Activity	XP
Daily login	5
Quiz completion	20

Activity	XP
90% score	40
Flashcard review	10

19. Student Dashboard Architecture

Dashboard Sections

Hero Section

- Student level
 - XP
 - Rank
 - Streak
-

Learning Section

- Pending teacher-assigned quizzes
 - Global practice quizzes
 - Global flashcards
 - Continue learning
 - Recent activities
-

Analytics Section

- Subject performance
 - Weekly progress
 - Accuracy graph
-

Leaderboard Section

- Rank
 - Top students
 - Quiz rankings
-

20. Teacher Dashboard

Features

Academic Controls

- Generate personalized quizzes
 - Generate personalized flashcards
 - Assign quizzes
 - Assign flashcards
 - Create remedial practice
-

Analytics

- Student performance
 - Weak topics
 - Quiz completion rate
 - Average class score
-

21. School Admin Dashboard

Features

School Statistics

- Total students
 - Total teachers
 - Active quizzes
 - Engagement rate
-

Monitoring

- Academic performance
 - Student activity
 - Teacher activity
-

22. Database Design

Main Tables

Authentication

- users
 - roles
 - schools
-

Academic Structure

- classes
 - sections
 - subjects
 - teacher_mappings
 - student_mappings
-

AI Content

- uploaded_documents
 - document_chunks
 - flashcards
 - quizzes
 - quiz_questions
-

Student Activities

- assignments
 - quiz_attempts
 - student_answers
-

Gamification

- student_xp
 - streaks
 - badges
 - leaderboard_entries
-

23. Backend API Architecture

Authentication APIs

Method	Endpoint	Purpose
POST	/login	User login
POST	/register	User registration
POST	/reset-password	Reset password

Document APIs

Method	Endpoint
POST	/upload-document
GET	/documents

Flashcard APIs

Method	Endpoint
POST	/generate-flashcards
GET	/flashcards

Quiz APIs

Method	Endpoint
POST	/generate-quiz
POST	/assign-quiz
GET	/quiz/{id}

Student APIs

Method	Endpoint
POST	/submit-quiz
GET	/student-report
GET	/leaderboard

24. Recommended Project Folder Structure

```
backend/
|
├─ app/
|   ├── api/
|   ├── auth/
|   ├── models/
|   ├── services/
|   ├── rag/
|   ├── embeddings/
|   ├── quiz_generation/
|   ├── flashcards/
|   ├── analytics/
|   ├── gamification/
|   └─ database/
|
frontend/
|
├─ app/
├─ components/
├─ dashboard/
├─ teacher/
├─ student/
├─ admin/
└─ services/
```

25. Security Architecture

Security Features

- JWT authentication
 - Password hashing
 - RBAC validation
 - Secure APIs
 - File validation
 - School-wise data isolation
 - Protected routes
-

26. Scalability Considerations

Future Scalability

The architecture should support:

- Multiple schools
 - Thousands of students
 - Large document ingestion
 - Horizontal backend scaling
 - Cloud deployment
-

27. Future Enhancements

Phase 2 Features

- AI tutor chatbot
 - Voice-based learning
 - Multilingual support
 - Adaptive learning engine
 - Parent portal
 - Real-time competitions
 - AI-generated study plans
 - Personalized recommendations
-

28. Complete End-to-End Workflow

```
graph TD; A[School Admin creates school structure] --> B[Teachers upload NCERT/custom content]; B --> C[System processes documents]; C --> D[Chunks converted into embeddings]; D --> E[Embeddings stored in vector DB]; E --> F[Teacher generates flashcards/quizzes]; F --> G[AI uses RAG for contextual generation]; G --> H[Teacher assigns quizzes]; H --> I[Students attend quizzes]; I --> J[Auto evaluation performed]; J --> K[XP/streaks updated]; K --> L[Analytics dashboards updated];
```

29. Final Project Vision

The Student Academic Management System is designed as a modern AI-powered academic ecosystem that combines:

- Academic management
- AI learning assistance
- RAG-based content generation
- Gamified learning
- Educational analytics
- Personalized learning experiences

into one scalable platform for schools.

The system reduces manual academic workload while improving student engagement and educational quality using artificial intelligence.